Question for Practice

Principles Of Information Security - Spring 2024, IIIT Hyderabad

January 28, 2025

1. [10 points] Construct a PRNG assuming DLP is hard. What is a hardcore bit, and prove whether the MSB/LSB of the DLP problem is a hardcore bit.

2. [6 points] Define a OWF. What assumptions can you relax if you want to have symmetric key cryptography, but the adversary is a "Deterministic" Poly Time Turing Machine? [Medium]

3. [7 points] Define CPA security through a game. In CPA security, what happens if the adversary is given key to the encryption oracle instead of just the oracle access? (Hint: Kerchkoff's theorem)

4. [4 points] Define Negligible functions. Is the following equivalent to the definition of a negligible function? Explain why its different, or prove its the same.

$$\begin{aligned} f \text{ is negligible if } f(x) &= \frac{1}{g(x)} \\ \text{where } g(x) &= c.e^{poly(x)} \qquad c \in \mathbb{R}^+ \end{aligned} \tag{1}$$

5. [10 points] Are the following functions negligible

   - $f(x) = \dfrac{1}{x!}$
   - $f(x) = \dfrac{x}{2^x}$
   - $f(x) = \frac{1}{(\log x)!}$
   - $f(x) = \frac{1}{(\log \log x)!}$
   - $f(x) = \frac{2^{-n}}{1000}$

6. [7 points] Using your experience in security definitions, privide a definition for perfect pseudorandom generators $G : \{0,1\}^n \to \{0,1\}^{n+1}$. Furthermore, prove that such perfect PRGs do not exist.

7. [10 points] Assuming that DLP is hard in $\mathbb{Z}^*_{17}$(of course, it isn't really), using 4-bits to reperesent each of its elements, design a corresponding PRG $G : \{0,1\}^4 \to \{0,1\}^*$, and output the first six bits if seed is set to be the last 4 bits of the last 2 digits of your roll number.

8. [6 points] Prove that the shift cipher is perfectly secret as long as only one character in [a, ..., z] is encrypted

9. [4 points] For each of the following encryption schemes, state whether the scheme is perfectly secret. Justify your answer in each case.

   (a) The message space is $\mathcal{M} = \{0, \dots, 4\}$. Algorithm **Gen** chooses a uniform key from the key space $\{0, \dots, 5\}$. $\mathbf{Enc}_k(m)$ returns $[k + m \ mod\ 5]$, and $\mathbf{Dec}_k(c) = [c - k \ mod\ 5]$.

(b) The message space is $M = \{m \in \{0,1\}^l |$ the last bit of m is 0 $\}$. **Gen** chooses a uniform key from $\{0,1\}^{l-1}$. $\mathbf{Enc}_k(m) = m \oplus (k||0)$, and $\mathbf{Dec}_k(c) = c \oplus (k||0)$.

10. [2 points] If **(Gen, Enc, Dec)** is a perfectly secret encryption scheme with message space $\mathcal{M}$ and key space $\mathcal{K}$, then prove that $|\mathcal{K}| \geq |\mathcal{M}|$.

11. [10 points] Assume the adversary has the ability to obtain ciphertexts for arbitrary plaintexts, i.e., it can choose a message and receive an encryption of the message without knowing the secret key. Attacks that leverage this information are known as chosen plaintext attacks.

    (a) Using chosen plaintext attacks, show how to learn the secret key for the shift, substitution, and Vigenère ciphers. Your attacks should each use only a single plaintext. What is the smallest plaintext length that suffices for each attack?

    (b) For the Vigenère cipher, please consider two cases: (a) when the period t is known; (b) when the period t is unknown but an upper bound t_{max} on t is known. For the latter case, an asymptotic analysis of the required plaintext length suffices, i.e., an upper bound on the required plaintext length.

12. [10 points] PRG or not? Let $G : \{0,1\}^{2n} \to \{0,1\}^{2n+1}$ be a pseudorandom generator (PRG). For each part below, either prove or disprove that $G' : \{0,1\}^{2n} \to \{0,1\}^{2n+1}$ is necessarily a PRG no matter which PRG G is used.

    (a) $G'(x) := G(\pi(x))$ for, where $\pi : \{0,1\}^{2n} \to \{0,1\}^{2n}$ is any poly(n)-time computable bijective function. (You may not assume that π^{-1} is poly(n)-time computable.)

    (b) $G'(x||y) := G(x||x \oplus y)$, where $|x| = |y| = n$. (Note: Here, and throughout the course, $x||y$ refers to the concatenation of two strings x and y.)

    (c) $G'(x||y) := G(x||0^n) \oplus G(0^n||y)$, where $|x| = |y| = n$. (Note: Here, and throughout the course, 0^n and 1^n denote the string of 0s and 1s, respectively, of length n.)

    (d) $G'(x||y) := G(x||y) \oplus (x||0^{n+1})$, where $|x| = |y| = n$.

13. [6 points] Let $f : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be a pseudorandom function (PRF). For the functions f' below, either prove that f' is a PRF (for all choices of f), or prove that f' is not a PRF.

    (a) $f'_k(x) := f_k(0||x)||f_k(1||x)$

    (b) $f'_k(x) := f_k(0||x)||f_k(x||1)$

14. [9 points] Give complete details of how to construct Y from X in each of the following cases:

    (a) X = One-way permutation, Y = pseudorandom generator

    (b) X = pseudorandom generator, Y = one-way function

    (c) X = pseudorandom generator, Y = pseudorandom function

    For each construction, provide:

    - A detailed description of the construction
    - A proof sketch demonstrating that Y satisfies the required security properties given that X satisfies its security properties

15. [5 points] Let F' be a secure pseudorandom function (PRF). Construct a PRF F_a that satisfies both of the following properties:

(i) F_a is a secure PRF when the key is completely hidden

(ii) F_a becomes insecure when the first bit of the key is leaked, i.e., there exists an efficient adversary that can distinguish F_a from a random function with non-negligible advantage when given access to the first bit of the key

Your answer should include:

- A formal description of your construction of F_a using F'
- A proof that F_a is secure when the key is completely hidden
- A concrete attack demonstrating how F_a can be distinguished from a random function when the first bit of the key is known

16. [points] Let f be a negligible function. Consider the following definitions:

Definition 1 (Overwhelming Function) *A function f is overwhelming if $1 - f$ is negligible.*

Definition 2 (Noticeable Function) *A positive function f is noticeable if there exist a positive polynomial p and a number n_0 such that $f(n) \geq \frac{1}{p(n)}$ for all $n \geq n_0$.*

Now consider the function $Z : \mathbb{N} \to \mathbb{R}$ defined as:

$$Z(n) = \begin{cases} 1 & \text{if } n \text{ is even} \\ 2^{-n} & \text{if } n \text{ is odd} \end{cases}$$

Given this two definitions determine the nature of function Z

Justify your answer.

17. [10 points] In block cipher operations, we analyzed four basic modes (ECB, CBC, OFB, and Counter) with respect to how changing a single plaintext block affects ciphertext blocks. Now, let's examine the reverse scenario:

Consider a sequence of ciphertext blocks $c_1, c_2, \ldots, c_n$ where some ciphertext block c_j is erroneous $(1 \leq j < n)$. For each mode of operation, analyze how this error affects the decryption of the remaining blocks.

Specifically, determine which plaintext blocks among $x_j, x_{j+1}, x_{j+2}, \ldots, x_n$ are received correctly in each mode.

For the special case where c_1 is incorrect, determine the blocks that are decrypted incorrectly.

Justify your answer by explaining the error propagation characteristics of ECB mode.

18. [10 points] Let $G : \{0,1\}^n \to \{0,1\}^{n+1}$ be a secure pseudorandom generator (PRG).

1. Construct a new PRG $\hat{G}$ with expansion factor $poly(n)$ where $poly(n)$ is a polynomial in n
2. Prove that your construction $\hat{G}$ is secure, assuming G is secure
3. Calculate the exact expansion factor of your construction
4. Show that your construction is efficient (runs in polynomial time)

Note: Your construction should be general enough to work for any polynomial $poly(n)$. You may use G as a building block multiple times in your construction.

Hint: Consider how you can iteratively apply G to achieve larger expansion factors while maintaining security.

19. [10 points] Consider the following statement about encryption schemes:

> Let $(\mathsf{Enc}, \mathsf{Dec})$ be a fixed-length private-key encryption scheme for messages of length ℓ that is EAV-secure. Let $\mathcal{A}$ be a PPT algorithm, $\mathcal{D}$ be a distribution over $\{0,1\}^\ell$, and $f : \{0,1\}^\ell \to \{0,1\}$ be any function.

1. Prove or disprove that for any such $\mathcal{A}$, there exists a PPT algorithm $\mathcal{A}'$ and a negligible function negl such that:

$$\left|\Pr[\mathcal{A}(1^n, \mathsf{Enc}_k(m)) = f(m)] - \Pr[\mathcal{A}'(1^n) = f(m)]\right| \leq \mathsf{negl}(n)$$

where:

- The first probability is taken over:
  - choice of m according to $\mathcal{D}$
  - uniform choice of $k \in \{0,1\}^n$
  - the randomness of $\mathcal{A}$ and Enc
- The second probability is taken over:
  - choice of m according to $\mathcal{D}$
  - the randomness of $\mathcal{A}'$

2. If you proved the statement true, provide a construction for $\mathcal{A}'$. If you proved it false, provide a counterexample.
3. Explain the significance of this result in terms of the security properties of EAV-secure encryption schemes.

20. [3 points] Consider a stateful variant of CBC-mode encryption where the sender simply increments the IV by 1 each time a message is encrypted (rather than choosing IV at random each time). Show that the resulting scheme is not CPA-secure.

21. [1 point] Say CBC-mode encryption is used with a block cipher having a 256-bit key and 128-bit block length to encrypt a 1024-bit message. What is the length of the resulting ciphertext?

22. [2 points] Let negl_1 and negl_2 be negligible functions. Then, prove

1. *The function* $\text{negl}_3(n) = \text{negl}_1(n) + \text{negl}_2(n)$ *is negligible.*
2. *For any polynomial* p, *the function* $\text{negl}_4(n) = p(n) \cdot \text{negl}_1(n)$ *is negligible.*